



Institut Teknologi Bandung

Directorate of Sustainable Professional Education

MODULE

Large Language Models

Fine-tuning, retrieval, re-ranking, and guardrails — the practitioner's layer on top of chapters 15 and 16.

Duration 3 hours (2 sessions)

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Source Module material for Designing and Building AI Products and Services

Session Objectives

By the end of this session, participants can:

- 1 *Objectives pending — to be written by the module owner.*
- 2 Placeholder: explain the **pretraining / adaptation / retrieval** division of labour.
- 3 Placeholder: choose between **full fine-tuning, LoRA, and prompting** for a given problem.
- 4 Placeholder: build a **retrieval pipeline** with a re-ranker and measure it.
- 5 Placeholder: specify **guardrails** for a production deployment.

SITE

[Course home](#)

This is a template, not the module

This deck is scaffolding. The structure, theme, and rendering are fixed; the material is authored separately by the module owner and may be delivered as PDF or PPTX instead.

It exists so the course site has a real deck to link to from day one, and so that — if the module is authored here — every block kind it will plausibly need is already demonstrated and verified in both the LaTeX and the web output.

Slides carrying no final content are marked with an amber band, so that an unfinished deck **cannot be mistaken for a finished one** on screen.

On top of chapters 15 and 16

This module takes all of that as given and covers the **practitioner's layer**: what you do when the model is someone else's, the data is yours, and the answer has to be right.



Placeholder outline — replace with the module's own structure.

01

Pretraining and adaptation

Section lead pending.

Adaptation options, side by side

APPROACH	WHEN IT FITS	WHAT IT COSTS
Prompting	<i>pending</i>	<i>pending</i>
Few-shot prompting	<i>pending</i>	<i>pending</i>
LoRA fine-tuning	<i>pending</i>	<i>pending</i>
Full fine-tuning	<i>pending</i>	<i>pending</i>
Continued pretraining	<i>pending</i>	<i>pending</i>

Content pending. This slide is scaffolding — the structure is fixed, the material is authored separately by the module owner.

Code and output render like this

placeholder

PYTHON

```
import keras_hub
```

```
backbone = keras_hub.models.Backbone.from_preset("...")  
backbone.enable_lora(rank=8)
```

OUTPUT

```
Total params:      ...  
Trainable params:  ...  
Non-trainable params: ...
```

Program output uses the `out` block, which renders in a distinct panel so a result is never mistaken for source.

02

Retrieval and re-ranking

Section lead pending.

Where the re-ranker sits



The retriever optimises recall cheaply; the re-ranker optimises precision expensively, over far fewer candidates.

Content pending. This slide is scaffolding — the structure is fixed, the material is authored separately by the module owner.

Cards, stats, and quotes render like this

recall@k, pending

nDCG, pending

latency, pending

Dense retrieval

pending

Sparse retrieval

pending

Hybrid

pending

A pull quote renders like this — useful for a definition or a finding worth stopping on.

Attribution line

03

Guardrails and evaluation

Section lead pending.

What has to be checked, and where

- ① *Step one — pending.*
- ② *Step two — pending.*
- ③ *Step three — pending.*

Content pending. This slide is scaffolding — the structure is fixed, the material is authored separately by the module owner.

Numbered steps use the `steps` block; unordered points use `bullets`. Both wrap correctly in the LaTeX output at any length.

Summary slide — pending

- 1 *Takeaway one — pending.*
- 2 *Takeaway two — pending.*
- 3 *Takeaway three — pending.*

BACK

[Chapter 16 — Text generation](#)

NEXT

[Agentic AI](#)